

POLICY MEMO - FOR OFFICIAL USE

Why limiting climate change matters for Niger

To: Minister of Environment, Head of Niger’s UNFCCC delegation

From: Your name, Climate Adviser

Date: 14 October 2026

Re: What Niger stands to lose, and to gain if warming is limited

KEY MESSAGE

Heat is becoming the defining threat to Niger’s economy, which rests on outdoor, rain-fed farming and herding: almost 40% of GDP and more than 80% of jobs. Under current global policies, Niger would face about ?? days a year of dangerous heat by 2100, against ?? today; holding warming below 2 °C would limit this to about ?? days. Hotter air also dries soils and pastures, so the share of the country in severe drought rises from ??% to ??% under current policies. Niger emits a negligible share of global greenhouse gases yet faces some of the largest losses: pressing for a rapid global phase-out of emissions is a matter of national interest, not charity.

Why Niger is unusually exposed

- **An open-air economy.** Agriculture and livestock account for almost 40% of GDP and over 80% of jobs, and depend overwhelmingly on rainfall (World Bank, 2024).
- **Shocks are already frequent and costly.** The 2005 and 2010 droughts triggered national food crises; the 2024 floods killed 339 people and affected more than 1.1 million (AFP, 2024).
- **Demand is growing fast.** With one of the fastest-growing populations in the world, Niger needs more food, water and jobs every year, from a climate that makes each harder to secure.

Three priority impacts: below 2 °C versus current policies

Source: Climate Impact Explorer (Climate Analytics), Niger, annual, area-weighted average; NGFS “Below 2 °C” and “Current policies” scenarios (the latter leads to roughly 3 °C of global warming by 2100). Median of the models; model range in brackets.

Impact (CIE indicator)	Today	2100: Below 2 °C	2100: Current policies	Why it matters for Niger
1. Dangerous heat Days per year with heat index above 40 °C	?? days	?? [??–??]	?? [??–??]	Above this threshold, sustained work outdoors risks heat exhaustion and heat stroke. Most Nigeriens farm, herd or build outdoors, mostly without electricity or cooling. WASCAL researchers project that even at 1.5–2 °C of global warming, more than half the population of most of West Africa would face heat discomfort seasonally (Sylla et al., 2018); the deadly April 2024 Sahel heatwave would have been impossible without the warming observed so far (WWA, 2024).
2. Severe drought Share of land area with SPEI below –1.5 (%)	?? %	??% [??–??]	??% [??–??]	This index counts evaporation as well as rain, so it captures what heat does to soils and pastures. Millet and sorghum feed the country. WASCAL crop modelling for three Nigerien zones projects a steady decline in millet yields in most scenarios, worst under very high emissions late in the century (Mohamed et al., 2025); beyond about +2 °C of local warming, no plausible rise in rainfall offsets heat losses (Sultan et al., 2013). A drought year means higher grain prices, malnutrition and distress sales of livestock.
3. Extreme rainfall Annual maximum 5-day precipitation (mm)	?? mm	?? [??–??]	?? [??–??]	Extreme Sahelian storms have tripled in frequency since 1982 as land temperatures rose (Taylor et al., 2017), and climate change made the deadly 2022 rains in Niger, Nigeria and Chad about 80 times more likely (WWA, 2022). Downpours on bare, crusted soils flood Niamey and the Niger River valley, where people, infrastructure and irrigated rice are concentrated. Niger must prepare for droughts and floods.

Save your Climate Impact Explorer screenshot as cie_figure.png

Niger – days per year with dangerous heat risk (HI > 40 °C) – NGFS below 2 °C vs. NGFS current policies

Figure 1. Days of dangerous heat per year in Niger: the gap between the two curves is what limiting warming saves. By 2100, current policies would bring about ?? dangerous-heat days a year, against ?? in a below-2 °C world and ?? today. The hatched bands show model uncertainty and do not overlap by the end of the century: the models disagree on the exact number of days, but not on the size of the difference between the two paths. Source: Climate Impact Explorer, Climate Analytics.

Implications for Niger's national interest

1. The cost is counted in GDP points. The World Bank (2022) estimates that by 2050 climate change would reduce Niger's annual GDP by 2.2% in a wet, optimistic scenario but by 11.9% in a dry, pessimistic one, the largest loss among the five G5 Sahel countries. These figures are probably *underestimates*: they leave out migration, conflict and ecosystem degradation.

2. The real danger is a cascade, not a single number. A hot, dry year → failed harvests and pastures → higher grain prices and forced sales of livestock → malnutrition, household debt, migration to cities and farmer–herder tensions → emergency spending that crowds out investment. Heat makes each step worse, and current policies make such years far more frequent.

3. Adaptation has limits that only global mitigation can push back. Irrigation, drought-tolerant seeds, early warning and crop insurance are essential but costly, and outdoor work cannot be “adapted” away from a heat index above 40 °C. Every tenth of a degree avoided lowers an adaptation bill that Niger cannot pay alone.

Recommended negotiating position. (i) Defend the 1.5 °C goal and press major emitters for NDCs consistent with a rapid fossil-fuel phase-out; (ii) tie Niger's support to scaled-up, accessible adaptation finance with simplified access for Least Developed Countries; (iii) use the Loss and Damage Fund for droughts and floods, in coalition with the African Group and the LDC group.

What the evidence does, and does not, say

Rising heat is the most robust signal: all models agree on its direction. **Rainfall is far less certain:** models disagree on whether the Sahel gets wetter or drier, so the drought and extreme-rainfall figures carry wider ranges, and some studies even project small millet yield gains from CO₂ fertilisation (GIZ/PIK, 2021). Conversely, observed storm intensification is faster than most models simulate (Taylor et al., 2017), so the extreme-rainfall figures may be conservative. The tool's only crop indicators (maize, rice, soy) were set aside because Niger's staples, millet and sorghum, are not modelled. These figures are projections, not forecasts, and they describe hazards only: Niger's high social vulnerability means actual impacts are likely to be larger than the physical numbers suggest.

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